

Lab 05 Thinking in Objects

Objective

In this lab, you will learn how to:

- Declare and create java classes with their data fields, constructors, setters and getters and other methods.
- Create an application class that uses the java classes by creating objects from them and calling their methods.
- Declare methods to deal with data fields of the class.
- Apply the aggregation relationship and recognize it.
- Declare static data fields and static method.

Activity Description

Note: Your project should contains **3 classes**(BookTest (main), Person and Book())

The Person Class:

Create a class **Person**:

- Two **private String instance** data filed named **fName** and **lName**.
- The Person class must have a **constructor** that initializes the instance variables.
- Provide a **set** and a **get** method for each data fields.
- Create a method named **toString()** that returns a string description for the Person object.

The Book Class:

Design a class named **Book** to represent the book information. The class contains:

- One **private String instance** data filed named **title**,
- One **private Person instance** data filed named **author** (**aggregation**).
- One **private int instance** data filed named **publishYear**.
- One **private static int** data filed named **count** initialize with **zero** –to store the total number of Book objects created.
- two **constructor** :
 - A constructor with **full arguments (title, fName, lName, publishYear)** - using **this** to reference the hidden data filed.
 - A constructor with **full arguments (title, Person object, publishYear)**- using **this** to reference the hidden data filed.
 - In both constructors, you should increase **count** data field by 1.
- Provide a **set** and a **get** method for first three data fields.
- Create a method named **toString()** that returns a string description for the book object.
- Create a **static** method named **getCount()** to return total number of Book instances.

The Application Class:

Write a test program named **BookTest**

- At the beginning of code print total number of books (**should print zero**) by call static method **getCount()**
- Creates **3 Book** objects **using both constructors**:
 - For the first object, use the **first** constructor to initialize the data fields ("java", "Ali", "Ahmed", 2008).
 - For the second object, read the information from user then use the **first** constructor to initialize the data fields ().
 - For the third object, use the **second** constructor to initialize the data fields ("java2", Person ("Mohammed","Ali") , 2018).
- Display the all books information by invoking their **toString()** method as shown in the
- Print **again** total number of books (**should print 3**)

Sample Output:

```
run:
the total number of books: 0
#####
Enter book title:
C++
Enter first and last name of author:
Hussam Saeed
Enter year:
2010
#####books information#####
java, Ali Ahmed, 2008
C++, Hussam Saeed, 2010
java2, Mohammed Ali, 2018
#####
the total number of books: 3
BUILD SUCCESSFUL (total time: 18 seconds)
```